Engineering One-pager

1-pager: Monetisation Data Service HLDD

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Team/Group: Revenue Platform
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Status: In review
Due: **Dec 12, 2025**

Stakeholders

Review scope	Status	Name *(reviewer's guide)*
AEWG	LGTM	Nicholas Preema
	Conditionally LGTM	Josh Devlin
Data Domain Sponsor	LGTM	Alex Hruska

Monetisation Domain Overview

This HLDD lists the initial, domain-aligned data services and Semantic Layer Subject Area that will be created as part of the warehouse data architecture that enables AI-Ready Data, a company goal for 2026.

The Monetisation Domain is composed of all data related to revenue events that users interact with Canva. It empowers Canva's monetisation engine and provides the data foundation for:

- **Subscription Lifecyle**: subscription provision, subscription offer, coupon, trial and churn rate
- **Pricing & Packaging**: Subscription plan price, feature bundle
- **Bililng & Invoicing**: Charge and invoice, Canva product billing information, tax
- **Checkout Platform**: Payment forms, payment experience

- **Payments and Gateways**: Payment success, authorisation rate, payment option, payment attempt, payment gateway fees
- **Revenue Fraud Operation:** Chargeback and dispute, fraud mitigation, fraud operation, fraud analysis
- **User Payout & Refund**: Subscription earning, user payout, customer refund
- **AI Credit**: AI quota, AI credit usage, unified credit model

Monetisation Data Services:

To provide these capabilities, the User Voice domain is structured into two Data Services and a Semantic Subject Area:

- **Monetisation Conformed Layer** - Provides reusable entity-aligned data products for Subscription Charge, Invoice, Payment, Payment Gateway, Checkout, Payout, Refund, Subscription, Subscription Plan, Subscription Price, AI Credit, Revenue Operation and offer & coupon.
- **Monetisation Metric Layer** - Delivers business reporting insight for
  - Subscription ARR
  - Trial and Acquisition
  - Payment success
  - Payment experience & checkout
  - Revenue Fraud
  - Unified Credit Model
- **Monetisation Semantic Layer -** A logical AI-ready interface to the Metric Layer

Ownership Model :

Area	Responsibility Scope	Owner(s)
Domain Data Sponsor	Overall domain ownership	Alex Hruska
Monetisation Conformed Layer	Design and maintenance of data models, including orchestration, infrastructure, and cross-domain coordination.	KW Kaihao Wang AG Aiden Guerin
Monetisation Metrics Layer	Design and maintenance of data models, including orchestration, infrastructure, and cross-domain coordination.	KW Kaihao Wang AG Aiden Guerin
Monetisation Metrics Layer	Metric definition and stakeholder coordination	Alex Hruska Ben LS Dhivyan Sureshkumar EL Elvis Leng ML Maisie Lee MB Matias Boero SK Shannon Kwan SS Srik Srinivasan AG Aiden Guerin KW Kaihao Wang
Monetisation Semantic layer	Design and maintenance of the semantic views	Alex Hruska Ben LS Dhivyan Sureshkumar EL Elvis Leng ML Maisie Lee MB Matias Boero SK Shannon Kwan SS Srik Srinivasan

		AG Aiden Guerin KW Kaihao Wang

Monetisation Conformed Layer

- Entity Catalog and Public Data Products

Entity / Entity Group	Purpose	Public interfaces/Existing models (Optional)
Subscription Charge	To track and manage the state of a user's subscription. A charge is a record when the user chooses to pay for a Canva product. • Link to the generated invoice • Creating user and brand • Charge amount	• dim_subscription_charge
Invoice	To represent the product contract associated with customer's purchase. • Owning entity (org/ brand) • Creating user and brand • Customer information • Invoice Provider • Invoice line item • Tax	• fact_invoice • fact_invoice_license_paid • finance_tax_report_united_state
Payment	To capture customer's payment process. A payment record could be represented as multiple attempts at processing through gateways. • Payer • Market reference • Mid • Merchant of record • Creating user and brand • Payment option • Payment risk assessment • Payment charge • Payment charge dispute	• fact_payment_attempt • fact_payment_attempt_invoice_link • fact_payment • dim_payment_option • fact_payment_fraud_blocked • fact_charge • fact_charge_dispute
Payment Gateway	To optimize payment success rate by routing different payment gateway providers for Canva user to process payment. • Gateway transaction • Gateway fee • Gateway dispute and chargeback	• fact_charges_stripe • fact_fees_stripe • fact_transactions_paypal • fact_transactions_ebanx • gateway_fees_ebanx • fact_transactions_adyen_payment • non_transactional_scheme_fees • dim_billing_address

Checkout	To record the customer checkout flows at Canva with available payment methods and trigger payment with collected information.
	
	● Payment forms:
	  ○ reached
	  ○ submitted
	  ○ submitted_failed
	  ○ submitted_succeeded
	● Payment method selected
	● External step initiated in checkout flow
		● fact_payment_experience
		● fact_payment_form_reached
		● fact_payment_form_submitted
		● fact_payment_form_submit_failed
		● fact_payment_form_submit_succeeded
		● fact_payment_method_selected
		● fact_payment_external_step_initiated

Payout	To capture the amount each brand has earned and payment made to contributors. ● License earning ● Subscription earning ● Supplementary earning ● Payout payment ● Payee dimension To capture the refund process performed internally or externally at canva.	● fact_payout_license_earning ● fact_payout_subscription_earning ● fact_payout_supplementary_earning ● fact_payout_payment ● dim_payee
Refund	● Refund charge record ● Refund invoice ● Refund reason ● Refund status ● Refund user and brand To manage the life cycle of a subscription that include both online (web & mobile) and offline (enterprise): ● Subscription lifecycle ● Subscription trial ● Subscription seat ● Subscription contract To capture the product features and configuration of a subscription	● fact_refund ● finance_tax_report_refund ● dim_fraud_brand ● fact_subscription_rollup ● fact_education_user_fraud_score ● fact_fraud_brand_details_by_date ● dim_subscription_plan ● ref_offline_subscription_plan
Subscription*		
Subscription Plan*		

Subscription Price*	To capture the pricing and billing intervals of subscriptions	• dim_subscription_plan_pricing
AI Credit	To capture Canva user's AI credit usage and infrastructure cost. • Canva user quota (entitlement) • User & brand capability • User quota tier	• fact_quota_transaction • fact_ai_credit_usage • dim_quota_feature • ref__quota_tier_limits • ref__quota_tier_mapping
Offer & Coupon	To provide Canva reward campaign reference • Offer • Coupon	• dim_coupon_campaign • fact_subscription_coupon_redemption • dim_offer_campaign • fact_offer_campaign_redemption

Note*: The subscription data is widely consumed internally in Canva and is modeled in the Foundation conformed layer.

- ## Conformed Layer Orchestration

Event Trigger Orchestration

- Triggered upon completion of **Foundation** Conformed layer processing in a daily basis:
  - Subscription Charge

Scheduled Cron Orchestration

- Runs on a fixed schedule (daily, not dependent on upstream completion)
  - Charge Invoice
  - Payment (Payment Gateway , Refund, Revenue Fraud, Checkout)
  - AI & Credit
  - Offer & Coupon

Entity Relationship in high-level

Monetisation Metrics Layer

- ## Metrics / Dimensions in Scope

Grouping	Metric / Dimension	Existing Models (Optional)
Revenue Platform	**Checkout Experience** - Monitor the proportion of users that have reached different steps among payment checkout forms that include: • Payment method selected rate • Payment form success rate • Payment form submit rate • Payment form submit success rate • Error recovery rate	• payment_experience_by_date • fact_payment_experience (for more dimensions)

Revenue Platform	**Payment Processing** - Monitor how efficient our payment platform has enabled the user to successfully make a payment:
	
	• Payment success rate
	  ○ Initial payment
	  ○ Renewal payment
	• Authorisation rate
		• fact_payment_attempt_invoice_link
		• invoice_summary_by_date
		• fact_payment_attempt

Supergroup	**Subscription Health** - % of cancelled paid subscriptions:
	
	• Paid subscription
	• Retention rate
	• Churn rate
	  ○ gross churn rate
	  ○ Delinquent churn rate
	  ○ Voluntary churn rate
		• dim_subscription

Revenue Platform	**Fraud, Disputes and Abuse** - Fraud related metrics during the user subscription lifecycle:: • Dispute rate • Dispute recovery rate • Chargeback loss rate • Payment loss fraud rate • Fraud warning rate • Payment form fraud block rate	
Revenue	**Customer Refund** - User	

Platform	outcomes and operational load indicators • Refund rate	
Revenue Platform	**Gateway Cost Efficiency** - Amount and percentage of fees charged by the payment gateway providers • % of fees as a share of total transaction volume • Gateway transaction success rate	
Revenue Platform	**Payout Platform** - The success and accuracy of payout payment: • Payout failure rate • % of earnings should be paid	
Growth	**Annualised Recurring Revenue (ARR)** - Annualised revenue from subscriptions, calculated by deriving from the monthly recurring revenue (**MRR**) • Subscription recurring revenue • Prepaid non-recurring revenue • average revenue per user/subscription	• fact_subscription_daily • subscription_arr_by_date • canva_arr_by_date • _stg_revenue_non_recurring_breakdown
Growth	**Trial and Acquisition** - User subscribe to Canva with/without trial: • CF2: MAU to trial • CF3: Trial to paid • Request to cancel • Trial abuse rate	• dim_subscription
Growth	**Seat Expansion** - To track brand size movement event	• fact_brand_invite_sent • fact_brand_invite_accepted • dim_team_invites_and_accepts • fact_team_admin_requests • dim_team_invite
Revenue	**Pricing Tier** - To monitor	

Platform	subscription pricing change and performance	
Revenue Platform	**Quota Service / Unified Credit Model (UCM)** - Monitor the AI credit and usage for Canva user to calculate the operation cost • feature tier • number of interactions and generations • feature cost • average AI feature cost per user	• `ai_credit_usage_by_month` • `fact_quota_transaction`
Revenue Platform	**Offer & Coupon performance** - Track coupon/offer campaign performance • offer/coupon redemption • coupon redemption rate	

- ## External Domain Dependencies

Owning Domain	Entities	Models in Transform (Optional)
Foundation	• User • Brand • Subscription • Country • Locale • Brand Size • Date Dimension • User Geography • Design metadata • Salesforce Account • Salesforce Opportunity • Salesforce Account Department • Salesforce Brand	• foundation.ref_currency_exchange_rate • foundation.dim_user • foundation.dim_brand • foundation.dim_subscription • foundation.fact_subscription_daily • foundation.fact_subscription_event • foundation.fact_subscription_history • foundation.fact_active_event_consolidated • foundation.fact_user_signup_event • foundation.dim_country • foundation.dim_locale • foundation.dim_brand_size • foundation.dim_date • foundation.fact_active_user_daily • foundation.dim_doctype • foundation.fact_design_open_event • foundation.dim_design • foundation.dim_brand_email_domain • foundation.dim_brand_membership • foundation.dim_brand_primary_account • foundation.dim_salesforce_account • foundation.dim_salesforce_account_department • foundation.dim_salesforce_account_related_brand • foundation.dim_salesforce_opportunity • foundation.dim_salesforce_brand • foundation.salesforce_account_activity_by_date
Growth	• Signup • Brand Movement	• fact_user_signup • fact_brand_user_activity_by_date
Sales & B2B	• Offline Seats • Offline ARR	• subscription_offline_arr_seats_by_date • salesforce_account_activity_by_date • subscription_offline_reporting_opportunity_cleaned
Ecosystem	• Appstore	• fact_appstore_app_used
Product Usage	• User Segment	• fact_ai_users_by_segments

		• dim_android_device • _stg_profile_brand • _stg_brand_owner • _stg_user_auto_signups
User Voice	• Specialist • Support Ticket	• dim_specialist • dim_support_ticket
MAU	• Monthly Active User	• fact_user_activity_status_daily • fact_managed_user_status_daily • maucast
Teams & Education	• Brand Membership	• dim_brand_membership • dim_brand_plan_history • fact_design_published • fact_design_shared • dim_education_user • dim_brand_user
Content	• CTA	• cta_source_classification
Discovery	• Creator	• _stg_profile_brand
Print	• Parternship Invoice	• _stg_print_partnership_invoice
Others		• netsuite__transactions

- ## Metrics Layer Orchestration

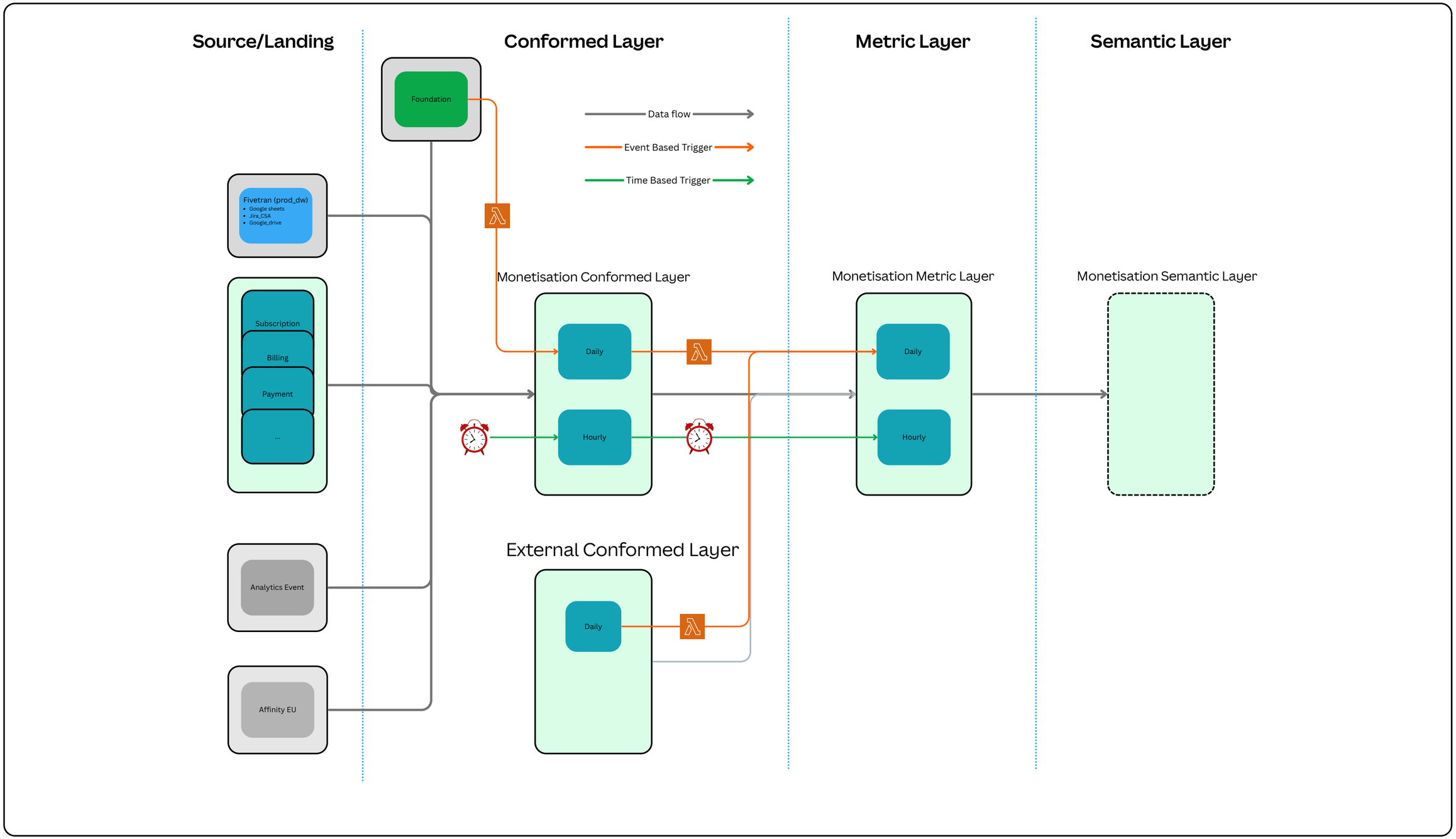
Event Trigger Orchestration

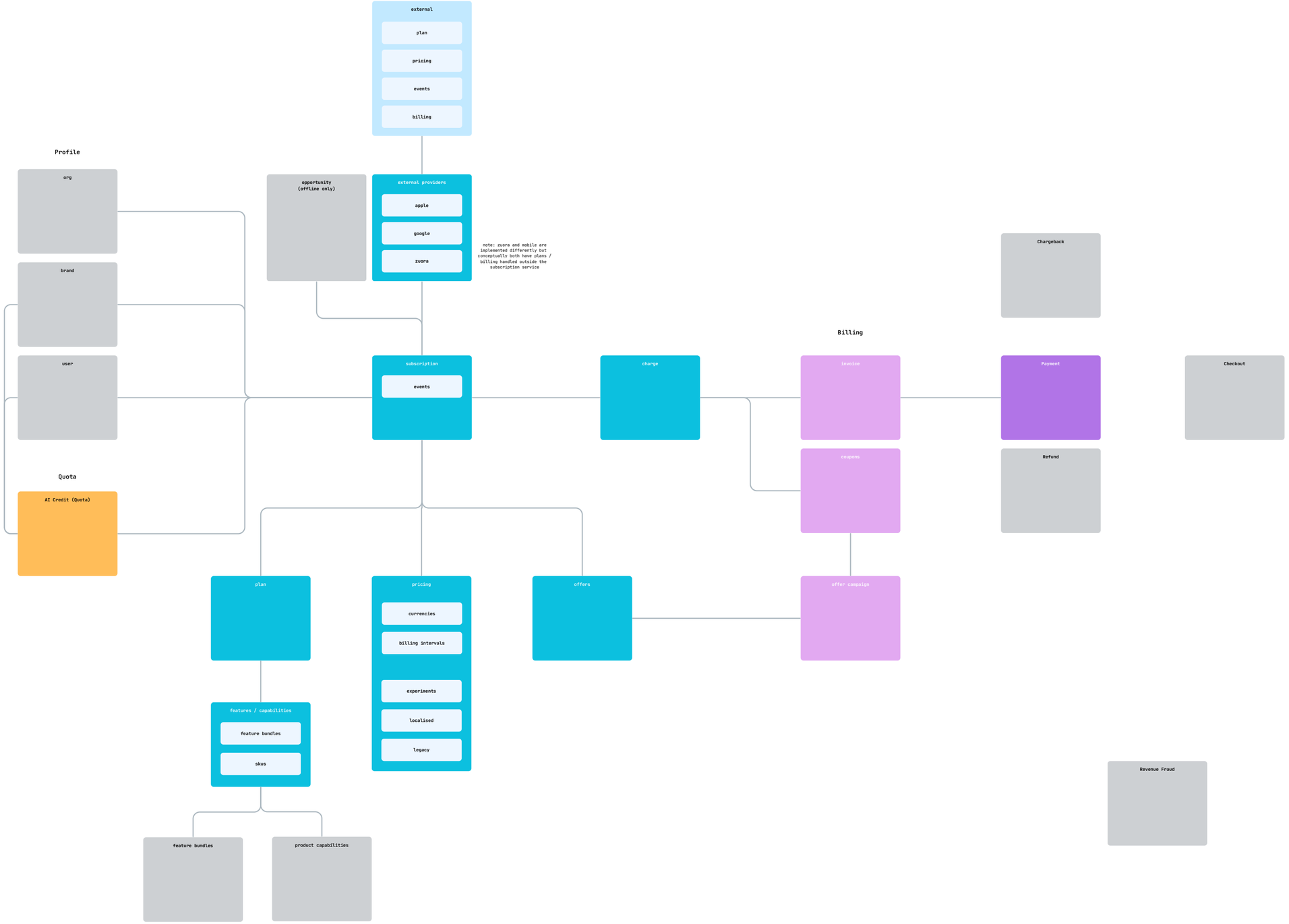
- Triggered upon completion of Foundation Conformed layer processing in a daily basis:
  - Subscription Health
  - Annualised Recurring Revenue (ARR)
  - Trial and Acquisition
  - Seat Expansion
  - Pricing Tier

Scheduled Cron Orchestration

- Runs on a fixed schedule (daily, not dependent on upstream completion)
  - Checkout Experience
  - Payment Processing
  - Fraud, Disputes and Abuse
  - Customer Refund
  - Gateway Cost Efficiency
  - Payout Platform
  - Quota Service / Unified Credit Model (UCM)
  - Offer & Coupon performance

Overall Orchestration


Source/Landing
Conformed Layer
Metric Layer
Semantic Layer
Foundation
Data flow
Event Based Trigger
Time Based Trigger
Fivetran (prod_dw)
Google sheets
Jira_CSA
Google_drive
Subscription
Billing
Payment
...
Analytics Event
Affinity EU
Monetisation Conformed Layer
Daily
Hourly
External Conformed Layer
Daily
Monetisation Metric Layer
Daily
Hourly
Monetisation Semantic Layer


external
plan
pricing
events
billing
Profile
org
brand
user
Quota
AI Credit (Quota)
opportunity
(offline only)
external providers
apple
google
zuora
note: zuora and mobile are implemented differently but conceptually both have plans / billing handled outside the subscription service
subscription
events
charge
Billing
invoice
coupons
offer campaign
Payment
Refund
Chargeback
Checkout
plan
features / capabilities
feature bundles
skus
feature bundles
product capabilities
pricing
currencies
billing intervals
experiments
localised
legacy
offers
Revenue Fraud